

MUHAMMED HAMMAD S

AI/ML Engineer | Generative AI Developer

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience engineering Retrieval-Augmented Generation (RAG) pipelines, AI-powered assistants, and full-stack intelligent applications. Skilled in Python, FastAPI, and LangChain, with practical depth across semantic search, vector databases, embeddings, and large language model (LLM) integration. Built production-oriented AI backends and CLI-driven RAG tools alongside deployed machine learning projects covering classification, clustering, and predictive modeling. Bachelor's degree candidate in Computer Science & Business Systems (expected 2027) with a strong foundation in architecting scalable, real-world intelligent systems.

TECHNICAL SKILLS

- Programming Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS, PHP, Java (Basic)
- Machine Learning: Machine Learning, Supervised Learning, Unsupervised Learning, Feature Engineering, Scikit-learn, Deep Learning, TensorFlow, Keras, Model Deployment, Model Evaluation
- Generative AI / LLM: Generative AI, Large Language Models (LLM), Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, Semantic Search, Embeddings, LLM Integration, Natural Language Processing (NLP)
- Vector Search / AI Tools: FAISS, ChromaDB, Whisper, tiktoken, Vector Databases
- Web / Backend Development: FastAPI, REST APIs, React.js, Next.js, Tailwind CSS, Streamlit, Software Development, Software Engineering
- Data Science / Analytics: Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Statistics, Data Visualization, Data Analysis, Power BI, Excel
- Developer Tools / Cloud: Git, GitHub, Docker, Netlify, Oracle Cloud Infrastructure (OCI), AWS, CI/CD, Agile, Version Control
- Core Competencies: Problem Solving, Communication, Team Collaboration, Project Management, Analytical Thinking, Attention to Detail, Time Management

FEATURED PROJECTS

Sentinel-AI | *Python, FastAPI, SQLAlchemy, OpenRouter, JWT Authentication, Docker*

- Engineered a production-oriented AI assistant backend using FastAPI and SQLAlchemy, with a real-time conversational interface powered by LLM integration via OpenRouter.
- Implemented secure JWT-based authentication and a persistent data layer to support multi-user conversational sessions.
- Containerized the application with Docker for reproducible deployment; incorporated a crisis-detection module to flag sensitive conversational content for user safety.

Advanced Video QA System | *Python, LangChain, FAISS, ChromaDB, Whisper, Streamlit*

- Engineered an end-to-end RAG pipeline converting video/audio into searchable knowledge: transcript extraction via Whisper, semantic chunking, and embedding generation.
- Integrated FAISS and ChromaDB for vector retrieval with LangChain-orchestrated LLM responses, enabling context-aware answers with timestamp-based video navigation.
- Built a Streamlit interface for natural language querying over video content, using semantic retrieval to ground responses in source material.

Dcobot-CLI | *Python, CLI, RAG, Embeddings, API Integration*

- Engineered a RAG-driven CLI pipeline that automates document ingestion, semantic chunking, and embedding generation from unstructured files.
- Designed document-processing workflows transforming raw files into retrieval-ready vector representations, with API integration for downstream RAG workflows.

Video Transcriber | *Python, Faster-Whisper, Streamlit*

- Developed a video/YouTube transcription application converting audio into text using Faster-Whisper for efficient speech-to-text processing.
- Built a Streamlit interface to handle the end-to-end transcription workflow from media input to formatted text output.

Terminal AI Assistant CLI | *Python, CLI, JWT Authentication, SQLite*

- Built a terminal-based AI assistant with secure JWT authentication and persistent conversation storage in SQLite.
- Implemented JSON incident mirroring, export, and backup functionality for reliable, auditable conversation history; included a crisis-detection feature for flagging sensitive input.

Image Toolkit Pro | *Python, PyInstaller, Tkinter/PIL*

- Developed a desktop image-processing application distributed as a standalone Windows executable via PyInstaller.

PROFESSIONAL EXPERIENCE

AI/ML Intern — ThirdVizion Labs

07/2026 – Present

- Working on machine learning projects involving practical application of AI/ML concepts and model-development workflows.

AI/ML Intern — LabMentix Pvt. Ltd.

07/2025 – 01/2026

- Built end-to-end ML pipelines covering data preprocessing, EDA, model training, hyperparameter tuning, and evaluation.
- Implemented supervised and unsupervised algorithms (classification, regression, clustering) on real-world datasets using Scikit-learn.

Data Science Intern — Saiket Systems

09/2025 – 10/2025

- Performed data analysis and visualization on real-world datasets to derive statistical insights for BI reporting.

Data Science Intern — GNCIPL

2025

- Worked on data preprocessing, exploratory data analysis (EDA), and data visualization using Python, Pandas, and NumPy in support of basic machine learning workflows.

Python Intern — Navodita Infotech

09/2025

- Developed Python programs leveraging OOP principles, scripting, and automation; built reusable utility modules.

Web Development Intern — CodeBind Technologies

07/2025

- Built responsive web pages using HTML, CSS, and JavaScript; assisted in backend development with PHP and MySQL.

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science & Business Systems (CSBS)

Expected Graduation: 2027

Kingston Engineering College, Katpadi, Tamil Nadu | CGPA: 7.5+

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- Oracle Cloud Infrastructure 2025 Certified Foundations Associate
- AWS × Deloitte Australia — Solutions Architecture Job Simulation
- Tata Group — GenAI Powered Data Analytics Job Simulation
- Deloitte Australia — Data Analytics Job Simulation
- Reliance Foundation — Data Science for Everyone
- Reliance Foundation — Python for Data Science
- AI/ML Internship Certificate — LabMentix Pvt. Ltd.

Additional certifications and credentials available on portfolio: hammad.dpdns.org

LANGUAGES

English, Tamil, Urdu, Hindi